

Course Code : ECT 355-4

ISJT/MW – 23 / 1811

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

SMART SENSORS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) Illustrate your answers wherever necessary with the help of neat sketches.
- (3) All questions are compulsory.

1.
 - (a) Illustrate transfer function that maps sensor input to its response. What approximations are considered when nonlinearity cannot be ignored ? 6(CO1)
 - (b) What is importance of understanding input and output impedances of a sensor circuit ? 4(CO1)
2.
 - (a) A disc type 1DB102K-EC thermistor has resistance of $1050\ \Omega$ at 27°C . Can you estimate the temperature when the resistance is $2330\ \Omega$. Assume $\beta = 3140$ at 27°C . 5(CO2)
 - (b) Summarize micromachining processes used in fabrication of micromechanical components. 5(CO4)
3.
 - (a) Analyze the fabrication technique that is useful in producing the sensor parts of heights ranging from micrometers to centimeters. 5(CO4)
 - (b) Convince the overall reasons for the creation of TEDS in IEEE 1451 standards and its benefits. 5(CO3)
4.
 - (a) What is purpose of IEEE 1451 standards for Smart Sensors ? Can you write a brief outline on IEEE 1451 standards ? 5(CO3)
 - (b) What is Sensor material ? Enlist five materials that can be used in sensor fabrication. 5(CO2)

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5. (a) Plastics are typically insulators. Justify briefly the use of plastic as a sensor material. 5(CO2)
- (b) Illustrate any one application of smart sensor vis-à-vis its equivalent base sensor. Write a part number of sensors mentioned. 5(CO5)
6. (a) How can you justify use of Smart Sensors in Smart cars, what compels industry to use smart sensors in this application in place of traditional sensors ? 5(CO5)
- (b) What properties a Smart Sensor must possess if it is to be used in automated consumer applications such as smart cars or smart homes ? 5(CO5)

